

IBONG TIRIRIT (PIPE III)

The measure of the ability of a coal to withstand exposure to the environment elements without excessive crumbling

- a. crumbling coefficient
- b. weatherability
- c. cohesiveness
- d. attrition coefficient

Answer: (B)

A type of coal that tends to break apart as it burns thereby exposing the unburned coal to the combustion air.

- a. loose coal
- b. free burning coal
- c. non-cohesive coal
- d. aggregate

Answer: (B)

A type of coal that produced a fused coal mass as it burns.

- a. plastic coal
- b. caking coal
- c. soft coal
- d. high-moisture coal

Answer: (B)

A combustible coal content that produces a major atmospheric pollutant consequently, it is important that coal has a low percentage of this component.

- a. nitrogen
- b. sulfur
- c. carbon
- d. hydrogen

Answer: (B)

The sum of the pressure head and velocity head.

- a. static head
- b. pressure head
- c. velocity head
- d. dynamic head

Answer: (D)

Which of the following statement is/are true regards to the first law of thermodynamics for a change in state of a control mass

- a. since heat and work are path function Q-W must also be a path function
- b. Q-W is an inexact differential
- c. A and B
- d. Q-W is a point function

Answer: (D)

Which of the following statements is/are true with regards to heat and work.

- a. heat and work are both transient phenomena
- b. both heat and work are boundary phenomena
- c. both heat and work are path function
- d. all of the above

Answer: (D)

The height of the surface of the water above the gage point

- a. pressure head
- b. velocity head
- c. dynamic head
- d. static head

Answer: (D)

The head required to produced a flow of the water.

- a. static head
- b. pressure head
- c. velocity head
- d. dynamic head

Answer: (C)

The dynamic pressure of the liquid at pump section less the corresponding to the temperature at the same point, converted to liquid.

- a. NPSH
- b. specific head
- c.pump operating head
- d. suction head

Answer: (A)

The algebraic difference of the discharge and suction heads.

- a. pump operating head
- b. TDH
- c. A and B
- d. NPSH

Answer: (C)

The transfer of the heat energy from exhaust gases to compressed air flowing between the compressor and combustion chamber.

- a. intercooling
- b. convection
- c. regeneration

d. economics

Answer :(C)

The removal of heat from compressed air between stages of compression.

- a. regeneration
- b. exhaustion
- c. heat evacuation
- d. intercooling

Answer:(D)

As a general rule, the cooling water side of the condenser should be schedule for inspection for every:

- a. month
- b. 3 months
- c. 6 months
- d. year

Answer: (B)

If the compressor runs continually, the cause might be the:

- A. high-pressure cutout switch is jammed shut
- B. low pressure switch is jammed shut
- C. thermal bulb is not operating properly
- D. scale trap is clogged

Answer: (B)

If the temperature in the icebox is too high, the trouble could be:

- A. a clogged scale trap
- B. air in the system
- C. automatic controls not functioning properly
- D. insufficient cooling water to the condenser

Answer: C

If frost forms on the cylinders, the cause would be:

- A. expansion valve not open wide enough
- B. charging valve open too wide
- C. expansion valve open too wide
- D. dehydrator not working properly

Answer: C

The most likely cause of high superheat would be:

- A. too much refrigerant
- B. expansion valve open too wide
- C. expansion valve closed in too much wide
- D. back-pressure valve set too high

Answer: (B)

Short-cycling means that the machine:

- A. grounds out frequently
- B. stops and starts frequently
- C. runs too fast
- D. runs too slow

Answer: B

If the cooling water to the condenser should suddenly fail:

- A. the solenoid valve will close
- B. the compressor will shut down

- C. the expansion valve will close
- D. an alarm will ring to notify the engineer

Answer: (B)

The memory lost when the operating power is removed

- a. power memory
- b. volatile memory
- c. initial memory
- d. quick data

Answer (B)

Which of the following regions in a multi-stage turbine where efficiency a highest?

- A. High pressure region above 200 psi
- B. Intermediate pressure region
- C. Low pressure most region
- D. A and B

Answer (B)

Calculate the use factor of a 135 MW plant with a load faceter of 0.8 and a peak load of 120 MW if its operation is limited to 8500 hrs in a year

- A 0.90
- B. 0.73
- C. 0.70
- D. 0.78

Answer (B)

Calculate the net radiation heat flux exchange of two bodies with surface temperature of 25 deg C. The radiation heat transfer coefficient is 7.0 W/m²-k

- A. 700 w/sq m

- B. 750 w/sq m
- C. 665 w/sq m
- D. 650 w/sq m

Answer C

Calculate the reserve factor for a power plant with a load factor of 0.7 average load of 50 MW and rated capacity of 80 MW

- a. 2
- b. 1.5
- c. 1.12
- d. 1.6

Answer C

Maximum demand of 30 MW and 10 MW are connected to a power plant with a peak load of 70 MW. What is the diversity factor of the plant?

- a. 0.50
- b. 0.70
- c. 0.65
- d. 0.86

Answer C

A gas in piston-cylinder set-up undergoes an expansion of such rate that $PV^{1.2} = \text{constant}$. Before the expansion process the pressure of the gas is 300 kpa and the volume is 0.50 m³. If the final volume of the gas is 0.1 m³, calculate the work done by the gas

- a. 9.71 KJ
- b. 11.0 KJ
- c. 12 KJ
- d. 8 KJ

Answer A

The phenomenon in which air enters a submerged suction pipe from the water surface is called a:

- a. whirlpool
- b. vacuum
- c. vortex
- d. thixotropic

Answer C

Where groundwater (wells and springs) or surface water (streams and lakes) are used for feedwater, water hardness of ____ and alkalinity of pH 10 to be pH should be considered

- a. 0-10 ppm
- b 10-20 ppm
- c 20-30 ppm
- d 30-40 ppm

Answer A

Which of the following is not a solid fuel?

- a. peat
- b. briquettes
- c. tar
- d. coke

Answer C

Canisters of helmets shall be removed immediately after having been used on sucking water from a public service main or private-use water system

- a. horizontal split case pump
- b. vertical shaft turbine pump
- c. booster pump
- d. submersible pump

answer C

A small fitting with a double offset, or shaped like the letter U with the ends turned out

- a. elbow tee
- b. expansion loop
- c. spigot joint
- d. cross-over

Answer d

The condition in which droplets of water are carried by steam in the boiler

- a. priming
- b. breeching
- c. carryover
- d. condensation

Answer a

At the start of the compression process of an air-standard diesel power cycle the pressure is 100kpa and the temp is 15 deg C. for a mean effective pressure of 1362 kpa and compression ratio of 18 the heat of combustion developed is 1800 kj/kg. what is the thermal efficiency of the cycle?

- a. 65 %
- b. 59 %
- c. 55%
- d. 68%

Answer b

A circular chip ($k = 150 \text{ w/mk}$) with a diameter $d = 10\text{mm}$ and thickness $t = 4 \text{ cm}$ inside diameter and 12 cm outer diameter. The inside wall temperature of the pipe is maintained at 500 deg C while

the outside surface of asbestos at 90 deg C. Calculate the heat loss per meter length

- a. 1265w/m
- b. 1500 w/m
- c 1400 w/m
- d. 1100 w/m

Answer a

Energy is always

- a. being produced
- b. being destroyed
- c. not changing
- d. none of the above

Answer b

A 5m thick flat plane ($k = 60 \text{ w/mk}$) is well insulated on its sides, while the top and bottom surfaces are maintained at 150 deg. Assuming a steady state heat conduction, calculate the temperature 4 m below the top surface.

- a. 100 deg c
- b. 120 deg c
- c. 94 deg c
- d. 80 deg c

Answer c

A solid of diameter $D = 0.5 \text{ m}$ and surface emissivity of 0.8 contains electrons that dissipate 100 W of heat. If it does not receive radiation from the surroundings, what is the surface temp?

- a. 300 deg C
- b. 280 deg C
- c. 230 deg C
- d. 270 deg C

Answer C

Which of the following is basically a jet engine that exhausts into a turbine generator?

- a. aeroderivative gas turbine
- b. brayton engine
- c. industrial gas turbine
- d. joule turbine

Answer A

it is referred to as the maximum continuous power available from a hydroelectric power plant even under the most adverse hydraulic condition

- a. primary power
- b. continuous base power
- c. continuous peak power
- d. firm power

Answer D

Energy destruction is a result of

- a. heat transfer
- b. work production
- c. process irreversibility
- d. none of the above

Answer c

Working fluids used in refrigeration cycle decreases in temperature as they pass through the throttle valve. The Joule- Thompson coefficient of these fluids is

- a. negative
- b. zero
- c. positive

d. none of the above

Answer C

The interaction that occurs between a system and its surroundings as the system executes a process, which is the result of the system being at the temperature different from the surrounding, is

- a. mass transfer
- b. heat transfer
- c. work transfer
- d. none of the above

Answer B

Working fluids used in refrigeration cycle decreases in temperature as the pass through the throttle valve. The joule-thompson coefficient of these fluids is

- a. negative
- b. zero
- c. positive
- d. none of the above

Answer C

The interactions that occurs between a system and its surroundings as a system executes a process, which is the result of the system being at temperature different from the surrounding is

- A. mass transfer
- b. heat transfer
- c. work transfer
- d. none of the above

Answer b

An air pollutant which is the result of burning fuels oils

- a. ferrous oxide
- b. hydrogen nitride
- c. sulfur dioxide
- d. aluminum oxide

Answer c

Analysis of solid fuel to determine moisture, volatile matter, fixed carbon, and ash expressed as percentage of the total weight of the sample

- a. viscometer
- b. proximate analysis
- c. ultimate
- d. pour point

Answer b

a gas analysis apparatus in which the percentage of oxygen, carbon dioxide and neonoxide in the gas measured by absorption in separate chemical solutions

- a. orsat
- b. proximate
- c. ultimate
- d. pour point

Answer a

the water added to boiler feed to compensate for the water lost through blowdown, leakage, etc

- a. condensate
- b. make-up
- c. sulfur
- d. economizer

Answer b

gas flow caused by boiler's pressure being less than boiler's furnace pressure

- a. induced draft
- b. infrared
- c. injector
- d. inhibitor

Answer (a)

An automatic pressure-relieving device that opens in proportion to the increase in pressure over its opening pressure

- a. pressure gauge
- b. relieve valve
- c. ammeter
- d. calorimeter

Answer b

A rotating device used to convert kinetic energy into mechanical energy

- a. turbine
- b. generator
- c. pump
- d. vacuum

Answer a

air can be prevented from getting into the system by:

- a. keeping all gland and stuffing boxes on the high pressure side tight
- b. keeping the dehydrator clean at all times
- c. keeping all glands and stuffing boxes on the low pressure side tight
- d. running the refrigerant through an aerator

Answer c

An excessively high head pressure could be caused by

- a. solenoid valve shut off
- b. insufficient cooling water to the condenser
- c. insufficient cooling water to the evaporator coils
- d. too much cooling water to the condenser

Answer b

The solenoid valve is controlled by

- a. the amount of liquid in the system
- b. the amount of gas in the system
- c. the temperature in the condenser
- d. the thermal switch operated by the icebox temperature

Answer D

if the discharge side of the thermal expansion valve is warmer than the

- a. expansion valve is working normally
- b. expansion valve is not working normally
- c. solenoid valve is not working normally
- d. scale trap is dirty

Answer B

If the pressure is too high

- a. the relief valve should open before the high pressure output
- b. the relief valve should open and let excess refrigerant go to receiver
- c. the high pressure output switch should operate before the relief valve
- d. close in on the suction valve

Answer c

If a pressurized water reactor (PWR)

- a. the coolant water is pressurized to work as moderator
- b. the coolant water boils in the core of the reactor
- c. the coolant water is pressurized to prevent boiling of water in the core
- d. no moderator is used

Answer c

Which of the following property of coal has major influence on the design and operation of a steam power plant

- a. density of coal
- b. carbon content of coal
- c. ash content of coal
- d. shape and size of coal

Answer C

a boiler horsepower is equal to the evaporation of 3405lbs of 100 degrees c water per hr. how much steam in lb per hour a 700 horsepower boiler can generate?

- a. 24150 lb / hr
- b. 21000 lb/ hr
- c. 18000 lb/hr
- d. 30000 lb/hr

Answer b

___ is obtained by mixing and burning precisely the right amount of fuel and oxygen so that products are left once combustion is complete

- a. perfect combustion
- b. burning combustion
- c. firing combustion

d. burning combustion

Answer a

what type of boiler commonly used in industrial and heavy commercial application due to less expensive to make low pressure, easier to maintain and repair

- a. water tube boiler
- b. fire tube boiler
- c. air tube boiler
- d. marine water

Answer b

during the emergency operation of the boiler, when there is no water in the gauge glass and steam comes out of the bottom trycock, the water level cannot be determined. What preventive measure the operator does to save the boiler

- a. shutdown the boiler and do not add water
- b. add water to compensate the water loss
- c. increase the firing rate
- d. continue to operate the boiler and add water

Answer a

the most important valve in the boiler operation and the last line of defense against disaster is known as

- a. pressure gauge
- b. compound gauge
- c. safety valve
- d. gate valve

Answer c

a common test and very important when a boiler is being repaired and or replaced before being back in operation or in service is known as

- a. hydrostatic test
- b. water level test
- c. test clamp
- d. acid test

Answer a

In heat transfer by conduction, due to symmetry the heat flow at a point is perpendicular to the isothermal surface through the point. This mode of conduction is a characteristic of

- a. non-isotropic
- b. isotropic
- c. insulators
- d. conductors

Answer a

why should you avoid bending or twisting of fan blades in an air conditioning unit?

- a. it will not cause ice build up
- b. it will wear out the motor bearings & cause noise
- c. it may not slice suction line
- d. it decrease the volume flow of refrigerant

Answer b

The heat transferred radially across insulated pipe per unit area

- a. remain constant
- b. inversely proportional to the thermal conductivity
- c. decrease from pipe wall to insulated surface
- d. increase from pipe wall to insulated surface

Answer c

under normal operating conditions, when a solid material to be dried is placed inside a batch tray dryer, the material is usually subjected

- a. under constant drying conditions
- b. under variable drying conditions
- c. under high temperature drying
- d. under low temperature drying

Answer a

a change of phase directly from vapor to solid without passing through the liquid state is called as

- a. sublimation
- b. vaporization
- c. solidification
- d. deposition

Answer d

in terms of heat transfer, the use of a cooling tower is said to be more efficient and more economical compared to an ordinary heat exchanger since

- a. large vol of air is available and free
- b. temperature profiles of air and water can cross each other
- c. large amount of water can be processed
- d. a cooling tower is much smaller and cheaper than a heat heat exchanger

Answer b

It is a form of oxygen photochemically produced in nature

- a. ozone
- b. oxidizing agent
- c. oxidation

d. D02

Answer a

What is the chemical formula of ozone

- a. O3
- b. D3O
- c. o7
- d. O1

Answer a

In the upper atmosphere, ozone is made by ultraviolet light reacting with

- a. oxygen
- b. nitrogen
- c. hydrogen
- d. water vapor

Answer a

The ozone concentration of parts per million (ppm) is generally considered maximum permissible for how many hours of exposure?

- A. 8 hrs
- b. 7 hrs
- c. 4 hrs
- d. 3hrs

Answer a

How may moisture be removed from air?

- a. condensation
- b. adsorption
- c. A and B
- d. none of these

Answer c

In cooling cycle, the dry bulb temp (db) of the air is lowered, what will happens to the relative humidity

- a. increased
- b. decreases
- c. remains constant
- d. increases or decreases depending on the temperature at which it is cooled

Answer A

What is the effect of superheating the refrigerant?

- a. it increases the coefficient of performance
- b. it decreases the coefficient of performance
- c. it lowers the boiling point of the refrigerant
- d. it increases the suction pressure of the refrigerant

Answer a

A turbine setting where one disadvantage is the difficulty of lubricating the operating mechanism, with consequent relatively rapid water

- a. siphon setting
- b. open fume setting
- c cast setting
- d. capacity setting

Answer b

A patented device combining expansion and compression function in a single rotor permitting higher cycle temperatures

- a. comprex

- b. compressor
- c. expander
- d. single-shaft

answer: A

A rough measure of the physical size of the equipment which must handle the specified quantity of the fluid

- a. pressure ratio
- b. fuel rate
- c. thermal efficiency
- d. air ratio

ans d

The ratio of the highest main compressor discharge pressure as the lower main compressor inlet pressure

- a, cycle pressure ratio
- b. cycle pressure level
- c. work ratio
- d. air ratio

ans a

The ratio of the maximum pressure in the cycle to the atmospheric pressure

- a. cycle pressure ratio
- b. cycle pressure level
- c. regenerator effectiveness
- d. thermal efficiency

ans b

The ratio of the actual partial pressure exerted by the water vapor in any volume of air to the partial pressure that would be exerted by the water vapor in the air is saturated at the temperature of the air

- a. relative humidity
- b. absolute humidity
- c. humidity ratio
- d. saturation ratio

ans a

In sensible heating the absolute humidity remains constant but the relative humidity

- a. increases
- b. remains constant
- c. decreases
- d. zero

ans c

The relative humidity becomes 100% and where the water vapor start to condense

- a. dew point
- b. saturated point
- c. cloud point
- d. critical point

ans a

Why should you avoid bending or twisting of fan blades in an air conditioning unit?

- a. It will cause ice build-up
- b. It will wear out the motor bearings and cause noise
- c. It may not slice suction line
- d. It decreases the volume flow of refrigerant

Answer: B

A type of refrigerant that will not damage the ozone layer.

- a. Hydrofluorocarbons (HFC's)
- b. R-22
- c. Hydrochlorofluorocarbons (HCFC's)
- d. R-12

Answer: A

Which of the following is the type of refrigerant that damages Ozone layer?

- a. Hydrochlorofluorocarbons (HCFC's)
- b. R-22
- c. R-12
- d. All of these

Answer: D

Large-bulb alcohol thermometer used to measure air speed or atmospheric condition by means of cooling.

- a. Kata thermometer
- b. JJ Thompson Thermometer
- c. Kelvin Thermometer
- d. Wet Bulb Thermometer

Answer: A

The phenomenon that warm air rise and cold air settle is called

- a. Stratification
- b. Setting Due
- c. Sedimentation
- d. Ventilation

Answer: A

Which of the following components of the window air conditioning system cleaned annually?

- a. Evaporator and condenser
- b. Fan blades and fan motor
- c. Motor and compressor
- d. All of these

Answer: C

A cooled heat exchanger that transfer heat from compressed air to cooler air.

- a. Intercooler
- b. Refrigerator
- c. Economizer
- d. Aftercooler

Answer: D

All the control stand, pitch of the turbine blades must be adjusted manually by a.

- a. Sensor
- b. Float valve
- c. Gate
- d. Switch

Answer: A

The most commonly used adjustable-blade propeller turbine.

- a. Francis turbine
- b. Kaplan turbine
- c. Impulse turbine
- d. Pelton wheel

Answer: B

Runner blades are adjusted automatically in synchronism with turbine wicker gates by a/an.

- a. Control valve
- b. Governor
- c. Oil servomotor
- d. Manual operation

Answer: C

What is supplied to the servomotor from the turbine governor oil system through the generator shaft and through a control valve.

- a. Water
- b. Air
- c. Steam
- d. Oil

Answer: D

Used to exhaust air from the casing for starting.

- a. Ejectors
- b. Dearator
- c. Injector
- d. Blower

Answer: A

Advantage for propeller turbines of all types

- a. open flumes setting
- b. siphon setting
- c. cost setting
- d. capacity setting

Answer: B

An office air conditioning system drew in a quantity of hot air from outside, cooled it to 1 deg C, and mixed it with the internal air of a building. Care was taken to supply enough fresh air to eliminate objectionable odors. The temperature of the space was maintained constant, and yet some people became uncomfortable in the afternoon, why?

- a. The relative humidity had dropped
- b. When a person spends a long time at a constant temperature that person becomes uncomfortable
- c. Radiant heat from others in the office made the space uncomfortable
- d. The complainers were not really uncomfortable, they were just bored

Answer: A

Equipment designed to reduce the amount of water vapor in the ambient atmosphere;

- a. Blower
- b. Condenser
- c. Dehumidifier
- d. Cooling tower

Answer: C

A large paper plant uses its own cogeneration power plant to generate its electricity because.

- a. It permits the use of a more efficient generator
- b. The company can better control the voltage
- c. It does away with high power lines leading to the plant
- d. It provides a more efficient conversion of fuel

Answer: D

The volumetric output of a centrifugal pump is reduced by closing a valve part way on the intake side of the pump, throttling the intake.

The reduction of flow is accompanied by “popping” noises which seem to come from inside the case.

What is the probable cause?

- a. The impeller has broken
- b. Dirt particles have got into the pump intake
- c. The intake valve flow seal is loose and vibrates
- d. The fluid is cavitating

Answer: D

If a man touches two metals which were kept together at room temperature, why would one metal feel colder than the other?

- a. One has a high temperature coefficient
- b. One has a lower temperature
- c. One has a high thermal conductivity
- d. One has a high temperature

Answer: C

The ratio of the internal thermal resistance of a solid to the boundary layer thermal resistance is described by:

- a. Biot number
- b. Nusselt number
- c. Prandtl number
- d. Reynolds number

Answer: A

An instrument which is commonly used to measure the most important characteristics of fuel oil which is viscosity is known as;

- a. Saybolt Viscosimeter
- b. Ph-meter
- c. Acid Testmeter
- d. Gravity meter

Answer: A

In a refrigerator, this is a chamber for storing low-side liquid refrigerant in a system, which also serves as a surge drum.

- a. Condenser
- b. Evaporator
- c. Compressor
- d. Accumulator

Answer: D

The measure of a fluid's resistance to flow (the internal friction of a fluid) as temperature increases.

- a. Density
- b. Viscosity
- c. Specific gravity
- d. Temperature

Answer: B

A mechanical device used to burn solid fuel specifically used in coal power plants.

- a. Stroker
- b. Boiler
- c. Stoker
- d. Condenser

Answer: A

The chemically correct ratio of air to fuel is a mixture capable of perfect combustion with no unused air or fuel.

- a. Fuel point
- b. Pour point
- c. Stoichiometric ratio
- d. Stack

Answer: C

When a solid material is said to be "dry" it means that

- a. The solid does not contain any moisture
- b. The solid still contains very small amount of moisture
- c. The solid contains equilibrium moisture
- d. The weight of the solid is the same as the bone dry weight

Answer: B

A hot block is cooled by blowing cool air over its top surface. The heat that is first transferred to the air layer close to the block is by conduction. It is eventually carried away from the surface by:

- a. Convection
- b. Radiation
- c. Conduction
- d. Thermal

Answer: A

When drying banana chips under the sun, the rate of drying is faster

- a. On a slightly breezy day
- b. On a calm day
- c. On a cloudy day
- d. On a hot and windy day

Answer: D

Reflector of a nuclear reactor are made up of

- a. Boron
- b. Cast iron
- c. Beryllium
- d. Steel

Answer: C

The function of a moderator in a nuclear reactor is

- a. To slow down the fast moving electrons
- b. To speed up the slow moving electrons
- c. To start the chain reaction
- d. To transfer the heat produced inside the reactor to a heat exchanger

Answer: A

Which of the following is the most commonly used moderator?

- a. Graphite
- b. Sodium
- c. Deuterium
- d. Heavy water

Answer: A

Ammonia leaks in the condenser can be detected by:

- a. Smelling the discharge water
- b. Applying litmus paper to the circulating water discharge
- c. Adding oil of peppermint to the system and tracing the smell
- d. Applying a soapy mixture to the condenser heads and looking for bubbles

Answer: B

Air arrangement of headwater tubes located in the boiler

- a. Economizer
- b. Condenser
- c. Evaporator
- d. Boiler

Answer: A

Scale in boiler can.

- a. Create low steam quality
- b. Cause foaming
- c. Over blow-off line
- d. Inhibit circulation and heat transfer

Answer: D

The effectiveness of a body as a thermal radiator at a given temperature.

- a. Absorptivity
- b. Emissivity
- c. Conductivity
- d. Reflectivity

Answer: B

In a cooling tower, the water is cooled mainly by;

- a. Condensation
- b. Evaporation
- c. Convection
- d. Conduction

Answer: C

A temperature above which a given gas cannot be liquefied;

- a. Cryogenic Temperature
- b. Absolute Temperature
- c. Vaporization
- d. Critical Temperature

Answer: D

What part of heat exchangers that are classified as radiant convection and placed at the near of the furnace, where they receive the bulk of the radiation from the flame.

- a. Radiant superheaters
- b. Condenser
- c. Radiator
- d. Evaporators

Answer: A

Which of the following is the major consideration while selecting the nuclear power plant?

- a. Safety
- b. Load center
- c. Disposal of waste
- d. Availability of water

Answer: C

A moderator is used to:

- a. Slow down neutrons
- b. Slow down electrons
- c. Accelerate neutrons
- d. Direct flow of electron

Answer: A

Which of the following material is not used as moderator?

- a. Graphite
- b. Beryllium
- c. Aluminum
- d. Light water

Answer: C

In nuclear power plant which of the following is used to slow down the fast neutron?

- a. Reflector
- b. Shield
- c. Control rod
- d. Moderator

Answer: D

Control rods for nuclear reactors are made of

- a. Graphite
- b. Concrete
- c. Cadmium
- d. Lead

Answer: C

The reactor in which fissionable fuel produced is more than the fuel consumed is known as

- a. Fast reactor
- b. Breeder reactor
- c. Thermal reactor
- d. None of the above

Answer: B

Heavy duty gas turbines typically have

- a. Double shaft
- b. Triple shaft
- c. Single shaft
- d. Quadruple shaft

Answer: C

Which of the following statements is not a key feature of fourier's heat conduction.

- a. It is phenomenological
- b. It defines an important material property called thermal conductance
- c. It is vector expression indicating that the heat flux is normal and the direction of decreasing temperature.
- d. Assume that the thermal conductivity is constant with respect to temperature of the material

Answer: D

What is the color code of steam line in geothermal power plant?

- a. Green
- b. Blue
- c. Silver Gray
- d. Red

Answer: C

What is absorbed by sulphites in boiler water treatment?

- a. Carbon dioxide
- b. Oxygen
- c. Carbon monoxide
- d. Impurities settled in mud drums

Answer: D

A flow volume in which the fluids flow upward through a tapered tube lifting a shaped weight to a position where the upward force just balances to weight.

- a. Rotameter

- b. Thermometer
- c. Viscometer
- d. Ammeter

Answer: A

An instrument that register total pressure and static pressure in a used to determine velocity.

- a. Ammeter
- b. Rotameter
- c. Pitot tube
- d. Pour point

Answer: C

A heat engine whose thermodynamic efficiency is greater than that of a Carnot device using the same energy reservoir is known as:

- a. A perpetual machine of the zeroth kind
- b. A perpetual machine of the second kind
- c. A perpetual machine of the first kind
- d. A perpetual machine of the third kind

Answer: B

The degrees of power interval in 4-stroke cycle of a six-cylinder engine is

- a. 60 degrees
- b. 180 degrees
- c. 90 degrees
- d. 120 degrees

Answer: D

The portion of sunlight, rich in ultraviolet rays, which has a strong effect on photographic plate.

- a. Gamma rays
- b. X-rays
- c. Beta rays
- d. Actinic rays

Answer: D

Which of the following does not illustrate the effect of temperature or pressure on gas solubility?

- a. Air bubbles from on the sides of a warm glass of water
- b. Soda pop is bottle under pressure
- c. Boiling frees water gases
- d. Air is more humid on rainy days

Answer: D

Among the effluents that may be released from the hydrothermal reservoir which one is the most toxic?

- a. Carbon dioxide
- b. Hydrogen sulfide
- c. Methane
- d. Ammonia

Answer: B

Newton said that a projectile if given enough horizontal velocity will not fall to the earth. Instead, it moves along a path around the earth. The 1st artificial satellite that verified Newton's statement is

- a. Apollo 1
- b. Houston 1
- c. Sputnik 1
- d. None of these

Answer: C

A refrigerant control valve that maintains a constant pressure in a evaporator

- a. Bleed valve
- b. Angle valve
- c. Automatic expansion valve
- d. Ball check valve

Answer: C

A rigid contains gas a 100 kpa and 10 degrees C. the gas is heated until the temperature reaches 100 degree C. Calculate the work done by the system.

- a. 2
- b. 0
- c. 1
- d. 1.5

Answer: B

The compression ratio in an air-standard Otto cycle is 10. At the beginning of the compression stroke the pressure is 120 kpa and the temperature is 15 degree C. Calculate the pressure and the temperature at the end of the compression process.

- a. 4,000 kPa
- b. 3,500 kPa
- c. 3,014 kPa
- d. 3,200 kPa

Answer: C

Find the thermal efficiency of an air-standard Otto power cycle with a compression ratio of 7.

- a. 60%
- b. 50%
- c. 65%
- d. 54%

Answer: D

At the beginning of the compression stroke in an standard Otto cycle. The pressure is 100 kPa and the temperature is 15 degree C. the compression ratio is 8 and work is 1,017.7 kJ/kg. Determine the mean effective pressure.

- a. 1,406 kPa
- b. 1,500 kPa
- c. 1,350 kPa
- d. 1,550 kPa

Answer: A

The compression ratio in an air standard diesel cycle is 20. At the beginning of the compression process the pressure and the temperature are 0.12 Mpa and 15 degree C respectively. What is the pressure at the end of the compression process?

- a. 8,500 kPa
- b. 7,955 kPa
- c. 8,300 kPa
- d. 8,200 kPa

Answer: B

The accounting of all energy units involved in a system a cycle of an piece of equipment

- a. Thermodynamic solution
- b. Equilibrium condition
- c. Summation of energy units
- d. Heat Balance

Answer: D

It refers to any arrangement whereby steam is bled from a turbine more pressure for any purpose whatsoever

- a. Regeneration
- b. Extraction
- c. Bled-off cycle
- d. Regenerative extraction cycle

Answer: B

Any arrangement whereby steam is bled from a turbine for a purpose of thermal regeneration of the condensate to a temperature level approaching that of the boiler water.

- a. Regenerative
- b. Extraction cycle
- c. Bled-off steam cycle
- d. Regenerative extraction cycle

Answer: A

The steam extraction from the turbine

- a. Bled steam
- b. Extraction
- c. Extracted steam
- d. A and C

Answer: D

The point of particular pressure where the steam is extracted

- a. Bleeder point
- b. Extraction point
- c. Bleeder
- d. A and B

Answer: D

Vapor pumps that raise the pressure of the refrigeration to a level that permits sufficient heat rejection to the condensing medium.

- a. Fans
- b. Blower
- c. Compressor
- d. Suction

Answer: C

Low temperature heat exchanger that is used to absorb heat from the refrigerated space and transfer it to the refrigerant in its tubes so that it can be transported to another and released.

- a. Condenser
- b. Evaporator
- c. Capillary tube
- d. Compressor

Answer: B

The substance employed as cooling effect to absorb heat from the refrigerated space is called

- a. Condensate
- b. Feedwater
- c. Refrigerant
- d. Moisture

Answer: C

It is inversely proportional to the power required to grind the coal to certain fineness

- a. Pulverizing index
- b. Fireness index

- c. Grindability index
- d. Inverse power coefficient

Answer: C

The temperature where the ash becomes very plastic, somewhat below the melting point of the ash.

- a. Ash- softening temperature
- b. Ash- plasticity temperature
- c. Ash – fluidity temperature
- d. Ash – liquefaction temperature

Answer: A

The amount of chemical energy in a given mass or volume of fuel.

- a. Heating value
- b. Energy storage
- c. Calorific value
- d. A and C

Answer: D

The refrigerant volume flow rate at entrance at the entrance of compressor were obtained from a test on a twin cylinder, single acting 15 cm x 20 cm, 320 rpm compressor ammonia refrigerating plant to be 33 L/s. Determine the volumetric efficiency of the compressor.

- A. 77.65 %
- B. 87.6 %
- C. 97.6 TR
- D. 65.65 %

Answer: B

A twin cylinder ammonia compressor with volume displacement of 14,726 cm³ operates at 300 rpm. Condenser and evaporator pressure

are 1200 kPa and 227 kPa respectively. Specific volume of refrigerant at the entrance of compressor is 528.26 L/kg. Compression process is polytropic with $n = 1.2$ and clearance factor of compressor is 2 percent. Determine horsepower required.

- A. 60 hp
- B. 70 hp
- C. 80 hp
- D. 90 hp

Answer: C

Three thousand cubic feet per minute of air are circulated over an air-cooled condenser. If the load on the condenser is 64,800 Btu/hr, compute the temperature rise of the air passing over the condenser. Specific volume of standard air (13.34 ft³/lb).

- A. 10 °F
- B. 15 °F
- C. 20 °F
- D. 25 °F

Answer: C

A reversed Carnot cycle has a refrigerating COP of 2.5. Determine the ratio T_H/T_L ?

- A. 1.4
- B. 1.5
- C. 1.25
- D. 1.2

Answer: A

Saturated vapor ammonia at -16 °C ($h_7 = 1442.60$ kJ/kg) leaves the evaporator and enters the compressor at -6 °C ($h_1 = 1465$ kJ/kg). The refrigerant leaves the condenser as saturated liquid at 40 °C ($h_4 =$

390.6 kJ/kg). Heat rejected from the condenser amount to 50 kW. The work to compressor is 208 kJ/kg, while the heat loss from compressor is 33 kJ/kg. If 9.5 kJ/kg of heat are lost in the piping between the compressor discharge and condenser inlet, determine the refrigerating capacity of the system.

- A. 49.5 TR
- B. 46.61 TR
- C. 12.88 TR
- D. 13.24 TR

Answer: A

A 35.0 mL sample of gas is enclosed in a flask at 22 degrees Celsius. If the flask was placed in an ice bath at 0 degrees Celsius, what would the new gas volume be if the pressure is held constant?

- A. 34.1 ml
- B. 31.1 ml
- C. 32.39 ml
- D. 33.1 ml

Answer: C

A car engine with a power output of 65 hp has a thermal efficiency of 24%. Determine the fuel consumption rate of this car if the fuel has heating value of 19,000 Btu/lb_m.

- A. 36.28 lb/hr
- B. 37.28 lb/hr
- C. 37.28 lb/hr
- D. 35.30 lb/hr

Answer: A

The thermal efficiency of a Carnot cycle operating between 170 °C and 620 °C to:

- A. 44%

- B. 50%
- C. 63%
- D. 73%

Answer: B

Compute the humidity ratio of air at 70% relative humidity and 25 °C when the barometric pressure is 101.325 kPa. From steam tables: $P_{\text{sat}} 34\text{ }^{\circ}\text{C} = 3.169\text{ kPa}$.

- A. 0.014 kg water vapor/kg dry air
- B. 0.14 kg water vapor/kg dry air
- C. 1.4 kg water vapor/kg dry air
- D. 0.0014 kg water vapor/kg dry air

Answer: A

A pressure gage registers 50 psig in a region where the barometer reads 14.8 psia. Find the absolute pressure in kPa.

- A. 666.66 kPa
- B. 556.66 kPa
- C. 446.66 kPa
- D. 336.66 kPa

Answer: C

Determine the quality of steam in a vessel containing 2 kg of saturated vapor and 8 kg of saturated liquid.

- A. 100%
- B. 20%
- C. 80%
- D. 60%

Answer: B

Consider 1 kg of air at 32 °C that expanded by a reversible polytropic process with $n = 1.25$ until the pressure is halved:

Determine the heat transfer. Specific heat at constant volume for air is 0.1786 kJ/kg.K.

- A. 17.02 kJ heat rejected
- B. 17.02 kJ added
- C. 7.05 kJ heat rejected
- D. 7.05 kJ added

Answer: B

Fifty kilograms of cooling water per second enter the condenser at 25 °C and leaves at 50 °C. Find the heat carried away by water.

- A. 1234.45 kW
- B. 5233.75kW
- C. 2340.52 kW
- D. 3140.25 kW

Answer: B

Ten kilograms per second of steam enter the turbine with an enthalpy of 3200 kJ/kg and enter the condenser with an enthalpy of 2500 kJ/kg in Rankine cycle. If the turbine efficiency is 80% and the generator efficiency is 90%, determine the power plant output.

- A. 4320 kW
- B. 3213 kW
- C. 4056 kW
- D. 5040 kW

Answer: D

The condenser of reheat power plant rejects heat at the rate of 500 kW. The mass flow rate of cooling water is 5 kg/s and the inlet cooling water temperature is 35 °C. Calculate the condenser cooling water exit temperature.

- A. 43.35 °C

- B. 53.45 °C
- C. 63.66 °C
- D. 74.34 °C

Answer: C

One kilogram of air is compressed adiabatically and in a steady-flow manner. The compression efficiency is 80% and the work done on the air is 265kJ/kg. Compute the heat.

- A. 212 kJ/kg
- B. 100 kJ/kg
- C. 0 kJ/kg
- D. 331.25 kJ/kg

Answer: C

Three-hundred kilojoules of heat flow by conduction from the outside to the inside of a cold storage in one hour. If the temperature and all other conditions are the same, what is the heat flowing through the cold storage room in two hours?

- A. 600 kJ
- B. 900kJ
- C. 300kJ
- D. 1,200 kJ

Answer: A

Determine the specific weight of air at 760 mmHg absolute and 22 °C?

- A. 3.517 kW
- B. 12,000 kW
- C. 211 kJ/min
- D. All of the above

Answer: D

Find the work process for a Helium gas at 20 °C

- A. 609 kJ/kg
- B. 168 kJ/kg
- C. 229 kJ/kg
- D. 339 kJ/kg

Answer: A

Two kilogram of gas is confined in a 1m³ tank at 200 kPa and 88 °C. What type of gas is in the tank?

- A. Helium
- B. Ethane
- C. Methane
- D. Ethane

Answer: B

Find the enthalpy of Helium if its internal energy is 200kJ/kg

- A. 144 kJ/kg
- B. 223.42 kJ/kg
- C. 333.42 kJ/kg
- D. 168 kJ/kg

Answer: C

Compute the mass of a 2 m³ propane at 280 kpa and 40 °C.

- A. 6.47 kg
- B. 5.1 kg
- C. 10.20 kg
- D. 9.47 kg

Answer: D

Compute the air flow in ft^3/min of mechanical ventilation required to exhaust an accumulation of refrigerant due to leaks of the system capable of revolving air from the machinery room for a mass of 4lbs refrigerant.

- A. 200
- B. 210
- C. 220
- D. 230

Answer: A

Compute the free aperture cross section in m^2 for the ventilation of a machinery room if the mass of refrigerant IS 9 KG.

- A. 0.314
- B. 0.414
- C. 0.514
- D. 0.614

Answer: B

A 29.53" X 39.37" pressure vessel contains ammonia with $f=0.041$. Compute the minimum required discharge capacity of the relief device in kg/hr .

- A. 106.71 kg/hr
- B. 108.71 kg/hr
- C. 110.71 kg/hr
- D. 112.71 kg/hr

Answer: C

Compute the maximum length of the discharge pipe installed on the outlet of a pressure-relief device in feet for internal pipe diameter of 0.5 in and rated discharge capacity is 8 lb/min of air. The rated pressure of relief valve is 16 psig.

- A. 0.286 ft

- B. 0.386 ft
- C. 0.486 ft
- D. 0.586 ft

Answer: A

A thermal power plant has a heat rate of 11, 363 Btu/kW-hr, Find the thermal efficiency of the plant.

- A. 28%
- B. 30%
- C. 34%
- D. 40%

Answer: B

What is the hydraulic gradient of a 1 mile, 17 inches inside diameter pipe when 3300 gal/min of water flow with $f = 0.03$

- A. 0.00714
- B. 0.00614
- C. 0.00234
- D. 0.00187

Answer: A

Find the loss of head in the pipe entrance if speed of flow is 10 m/s .

- A. 5.10
- B. 10.2
- C. 17.4
- D. 2.55

Answer: D

Find the mass of carbon dioxide having a pressure of 20 psia at 200F with 10 ft^3 volume.

- A. 1.04 lbs
- B. 1.14 lbs
- C. 1.24 lbs
- D. 1.34 lbs

Answer: C

Wet material, containing 220% moisture (dry basis) s to be dried at the rate of 1.5 kg/s in a continuous dryer to give a product containing 10% (dry basis). Find the moisture removed in kg/hr

- A. 3543.75 kg/hr
- B. 3513.75 kg/hr
- C. 3563.75 kg/hr
- D. 3593.75 kg/hr

Answer: A

Copra enters a dryer containing 70% moisture and leaves at 7% moisture. Find the moisture removed on each pound of solid in final product.

- A. 6.258 lb
- B. 1.258 lb
- C. 4.258 lb
- D. 2.258 lb

Answer: D

A 1 m x 1.5 m cylindrical tank is full of oil with S.G = 0.92. Find the force acting at the bottom of the tank in dynes.

- A. 106.33×10^3 dynes
- B. 106.33×10^4 dynes
- C. 106.33×10^5 dynes
- D. 106.33×10^6 dynes

Answer: D

Find the pressure at the 100 fathom depth of water in kpag.

- A. 1,793.96 kpag
- B. 1,893.96 kpag
- C. 1,993.96 kpag
- D. 1,693.96 kpag

Answer: A

Find the depth in furlong of the ocean (S.G = 1.03) if the pressure at the sea bed is 2,032.56 kpag.

- A. 1
- B. 2
- C. 3
- D. 4

Answer: A

Find the mass of 10 quartz of water.

- A. 10.46 kg
- B. 9.46 kg
- C. 11.46 kg
- D. 8.46 kg

Answer: B

Find the heat needed to raise the temperature of water from 30C to 100 C with 60% quality. Consider an atmospheric pressure of 101.325 kpa. Use the approximate enthalpy formula of liquid.

- A. 293.09 kJ/kg
- B. 1,772.90 kJ/kg
- C. 1,547.90 kJ/kg
- D. 1,647.29 kJ/kg

Answer: D

Find the enthalpy of water at 212F and 14.7 psi if the dryness factor is 30%. Use the approximate enthalpy formula of liquid

- A. 461 Btu/lb
- B. 471 Btu/lb
- C. 481 Btu/lb
- D. 491 Btu/lb

Answer: B

An air compressor consumed 1200 kW-hr per day of energy. The electric motor driving the compressor has an efficiency of 80%. If indicated power of the compressor is 34 kW, Find the mechanical efficiency of the compressor.

- A. 117.65%
- B. 75%
- C. 85%
- D. 90%

Answer: C

An air compressor has a power of 40 kW at 4% clearance. If clearance will increase to 7%, what is the new power?

- A. 70 kW
- B. 40 kW
- C. 53 kW
- D. 60 kW

Answer: B

A refrigeration system consumed 28,300 kW-hr per month of energy. There are 20% of energy is lost due to cooling system of the compressor and motor efficiency is 90%. If COP of the system is 6, Find the tons of refrigeration of the system.

- A. 43.15%
- B. 46.15%
- C. 49.15%

D. 41.15%

Answer: C

A 23 tons of refrigeration system has a heat rejected of 100kW. Find the energy efficiency ratio of the system.

- A. 13.42
- B. 14.42
- C. 15.42
- D. 16.42

Answer: B

A 200 mm X 250 mm, 8 cylinder, 4-stroke diesel engine has a brake power of 150kW. The mechanical efficiency is 80%. If two of the cylinders were accidentally cut off, what will be the new friction power?

- A. 31.50 kW
- B. 33.50kW
- C. 35.50 kW
- D. 37.50 kW

Answer: D

If the efficiency ratio of the refrigeration system is 12.6, what is the COP of the system?

- A. 3.69
- B. 4.23
- C. 5.92
- D. 6.83

Answer: A

What is the approximate value of temperature of water having enthalpy of 208 Btu/lb?

- A. 138.67°C

- B. 115.55°C
- C. 258.67°C
- D. 68.67°C

Answer: B

Convert 750°R to $^{\circ}\text{K}$

- A. 390.33°K
- B. 395.33°K
- C. 410.33°K
- D. 416.33°K

Answer: D

An otto cycle has a compression ratio of 8. Find the pressure ratio during compression.

- A. 18.38
- B. 16.38
- C. 14.38
- D. 12.38

Answer: A

A diesel cycle has a cut off ratio of 2.5 and expansion ratio of 4. Find the clearance of the cycle.

- A. 9.11 %
- B. 5.55%
- C. 11.11%
- D. 15.25%

Answer: C

A dual cycle has an initial temperature of 30°C . The compression ratio is 6 and heat addition at constant volume process is 600 Kj/kg . If cut-off ratio is 2.5, Find the maximum temperature of the cycle.

- A. 3638.5°C

- B. 3365.50°C
- C. 3565.50°C
- D. 3965.50°C

Answer: B

A three stages air compressor compresses air from 100 kPa to 1000 kPa . Find the intercooler pressure between the first and second stage.

- A. 505.44 kPa
- B. 108.44 kPa
- C. 316.23 kPa
- D. 215.44 kPa

Answer: D

A 10-stages air compressor compresses air from 100 kPa to 800 kPa . Find the intercooler pressure between the first and second stage.

- A. 282.82 kPa
- B. 113.21 kPa
- C. 123.11 kPa
- D. 333.51 kPa

Answer: C

A 3-stages air compressor compresses air from 100 kPa to 700 kPa . Find the intercooler pressure between the 2nd and 3rd stage

- A. 365.88 kW
- B. 375.88 kW
- C. 385.88 kW
- D. 395.88 kW

Answer: A

Carnot cycle A, B and C are connected in series so that the heat rejected from A will be the heat added to B and heat rejected from B

will be added to C. Each cycle operates between 30°C and 400°C . If heat added to A is 1000kW , find the work output of C.

- A. 111.44 kW
- B. 549.78 kW
- C. 247.53 kW
- D. 141.89 kW

Answer: A

